Lab 4
Localization using Particle Filter

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5/18/26

Trajectory Plot

The robot was driven in a full circle in Gazebo. The SIR particle filter (M = 500 particles) was fed wheel odometry for the predicted step and LIDAR detections of three known beacons for the weight/resample steps.

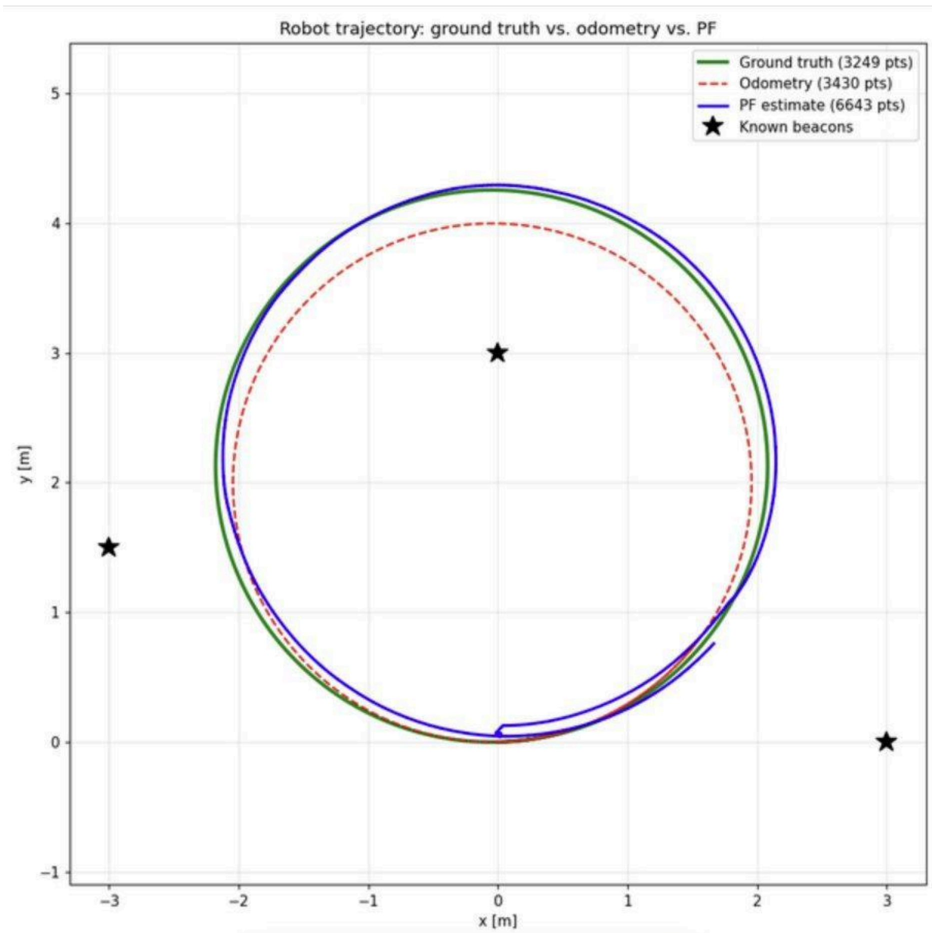


Figure 1: Ground truth (green), odometry (red dashed), particle filter estimate (blue).

Discussion Questions

Q1. Where does the PF estimate deviate most from ground truth, and why?

The most notable deviation of the PF estimate (blue) from the ground truth occurs in the first second or two at the origin. It is represented by the small loop and the jitter at the bottom of the circle. This is the beginning of the convergence phase. The particle cloud is spreading and has not collapsed to the true pose, and thus the weighted-mean estimate has not yet stabilized, and the estimate keeps drifting. As soon as the weighted mean is stabilized, blue and green paths coincide for the rest of the motion, with only a small amount of jitter due to resampling of the particle cloud. In addition, a moderately larger gap is observed to the right of the circle, which corresponds to the area where the robot is the farthest from the beacons, where the geometry of the situation provides the least amount of bearing information.

Q2. How long did the cloud take to collapse, and what was the limiting factor?

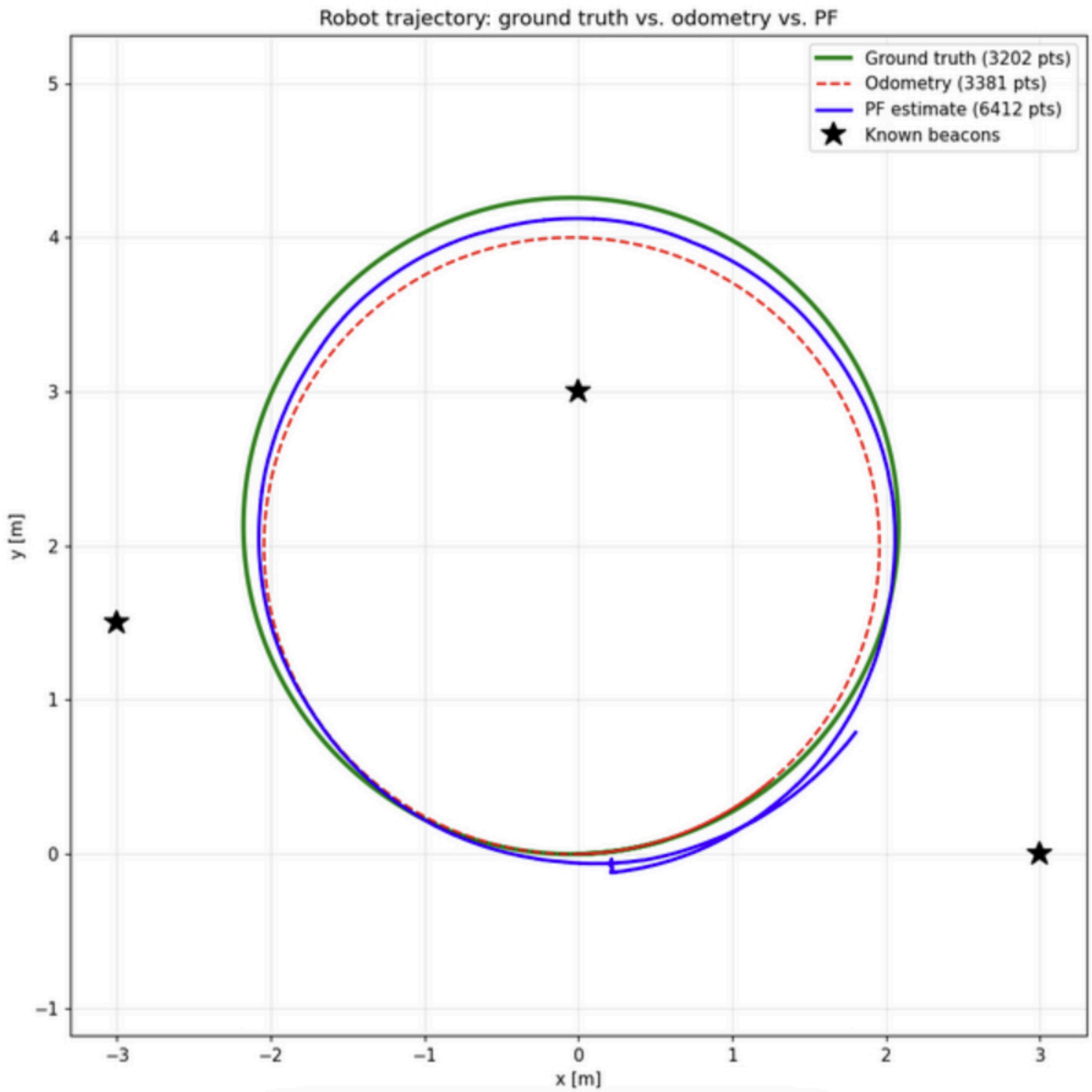
The cloud shrank around the robot in one to two seconds after the robot started moving. Due to the tight initial heading spread (INIT_THETA_SPREAD = 0.2 rad), the particles only needed to group along the x and y, further speeding up the dispersion. The rate of dispersion was affected by the number of beacons that could be seen; however, every LIDAR scan that picked up more than one beacon combined a number of likelihoods and instantly created a peaked weight distribution and concentrated particles. With only one beacon visible, the cloud would collapse much slower because one range-bearing measurement only slightly constrains position in comparison to two or three range-bearing measurements.

Q3. Compare the PF result to the UKF result from Lab 3 on the same circle

Once converged, both filters tracked ground truth well, but within the outer bounds of their convergence circles, the UKF produced a tighter and smoother estimate than the PF. This is expected. The UKF, which produces continuous estimates of state along the time domain by maintaining a single Gaussian (mean and covariance) of uncertainty and propagated through time by sigma points, produces estimates that are continuous and untainted by noise, except for updates from measurements. The PF, on the other hand, maintains a finite cloud of 500 samples to represent its belief and, therefore, its mean estimate is untainted by noise, but is still subject to jitter in its mean estimate due to a lack of sampling. The PF, however, drastically outperformed the UKF in convergence rate and reliability from a poor initial guess. This is because the PF can represent large uncertainty, due to the particles being spread throughout the region, and recover from poorly represented unconstrained initial uncertainty or even multi-modal ambiguity, whereas the UKF's single Gaussian representation will diverge if the initial mean is/was too far from the truth and the region of linearization contains the real state/pose. Thus, the UKF is more efficient when there is good initial guess, while the PF is more robust to poor initial guess and non-Gaussian uncertainty.

Q4. What happens with N_PARTICLES = 50 instead of 500?

Rerunning the trajectory with N_PARTICLES = 50 produced an almost identical trajectory plot to the run with N_PARTICLES = 500. The 50 particle version also produced a blue PF estimate that followed the ground truth very closely around the entire circle with only small amounts of jitter, especially during the first step of convergence to the origin, which was also slightly larger. For this problem, this shows that 50 particles was enough to indicate the belief well. Since the state is only three-dimensional and with three beacons providing frequent and strong measurements, a small number of particles is enough to capture the posterior accurately. A general particle count vs estimate smoothness balance still needs to be kept in mind. Numerous particles lead to a smooth, robust estimate with a high computational cost, whereas few particles lead to a rough estimate with a low computational cost. The value of an increased particle count would become especially clear in more challenging, higher dimensional states with sparse measurements or multi-modal uncertainty. In those cases, 50 particles would be too few to cover the belief and the estimate would degrade, losing track altogether.


Robot trajectory: ground truth vs. odometry vs. PF
Ground truth (3202 pts)
Odometry (3381 pts)
PF estimate (6412 pts)
Known beacons
y [m]
x [m]
5
4
3
2
1
0
−1
−3
−2
−1
0
1
2
3